

Q 1) Create Following webpage to demonstrate the various types of selectors in CSS :

CSS Selectors Demo

This is a paragraph styled using an element selector.

This is a heading styled using an element selector.

Another paragraph styled using the same selector.

Start of Div: This is a div styled using an element selector. - End of Div

Span: This is a span styled using an element selector.

Start of Div: This is a box styled using a class selector. - End of Div

Start of Div: Another box styled with the same class. - End of Div

This is a paragraph styled using a class selector.

Span: This is a span styled using a class selector.

This is a heading styled using a class selector.

Start of Div: This is a unique box styled using an ID selector. - End of Div

Start of Div: This is another box styled with a different ID selector. - End of Div

THIS IS A HEADING STYLED USING AN ID SELECTOR.

THIS IS A HEADING STYLED USING AN ID SELECTOR.

This paragraph is styled using an ID selector.

Span: This span is styled using an ID selector.

Start of Div:

This paragraph is inside a container styled using a descendant selector.

Span: This span is inside a container styled using a descendant selector.

Start of Div: This div is inside a container styled using a descendant selector. - End of Div

- This list item is inside a container styled using a descendant selector.
- Another list item inside a container.

- End of Div

Start of Div:

This is a child of the parent div styled using a child selector.

Span: This is another child of the parent div.

Start of Div: This div is also a child of the parent div. - End of Div

- This list item is a child of the ul in the parent div.

- End of Div

Adjacent Sibling Example 1

This paragraph is adjacent to the heading above it.

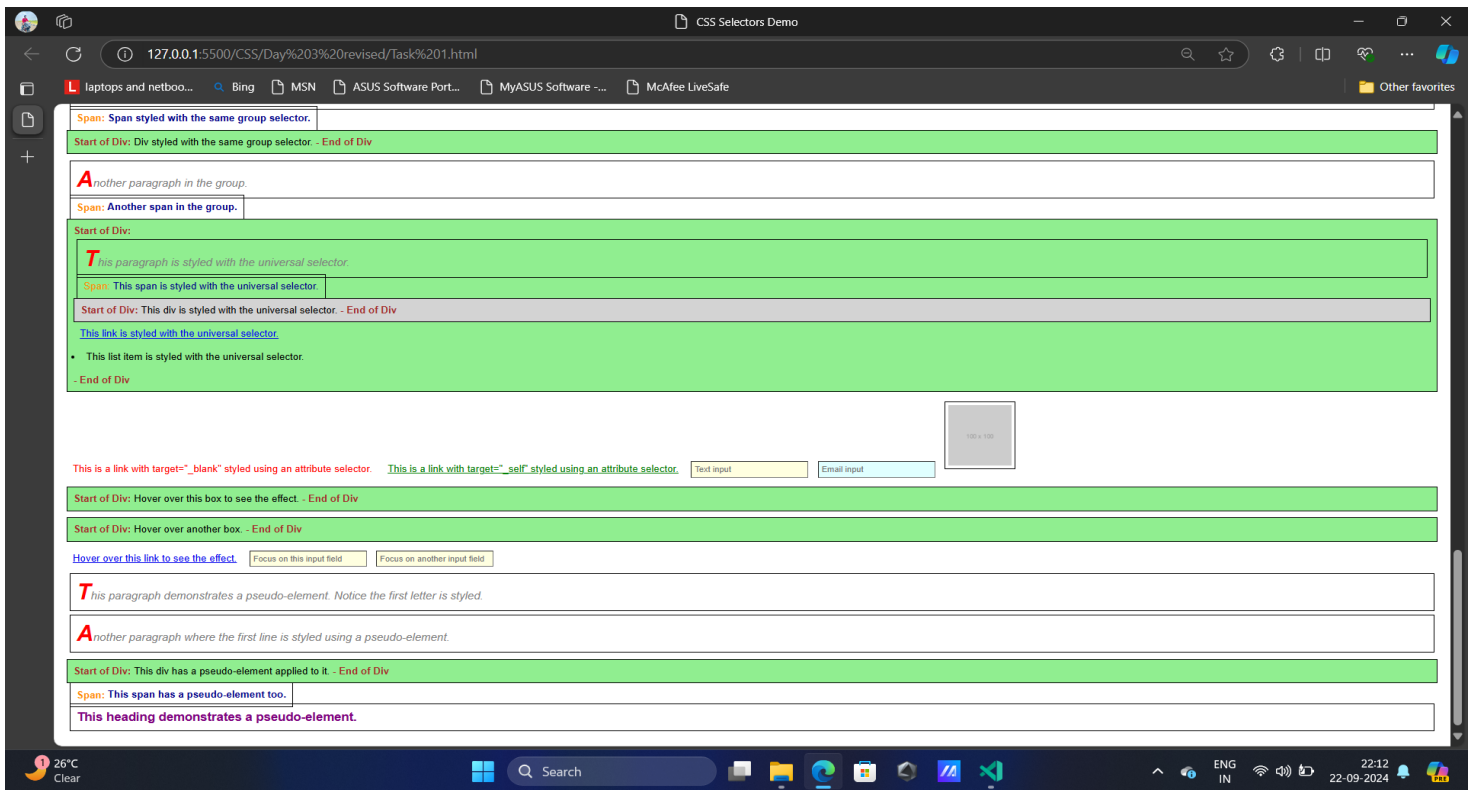
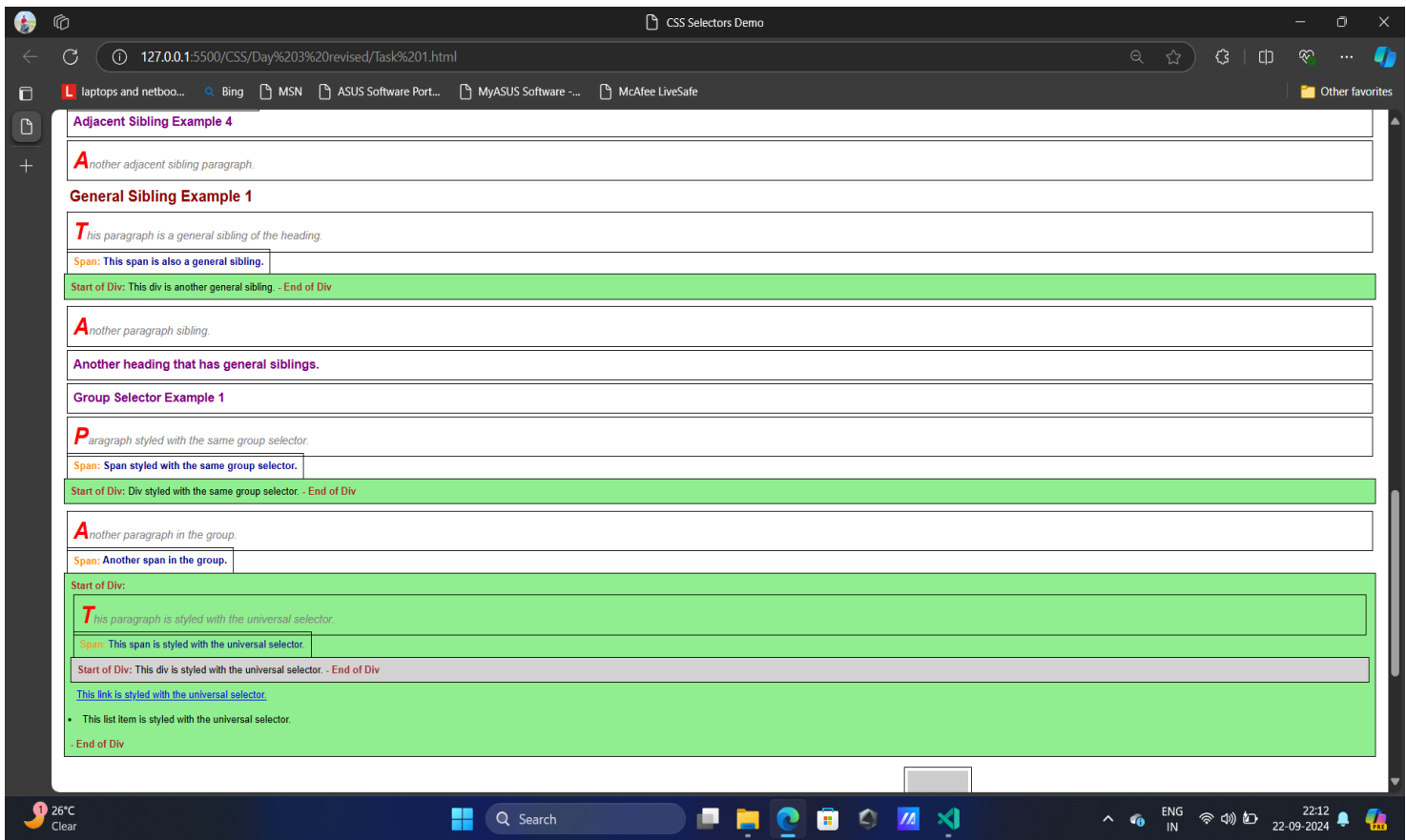
Adjacent Sibling Example 2

This is another adjacent paragraph.

Adjacent Sibling Example 3

Span: This is an adjacent sibling span.

Adjacent Sibling Example 4



```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>CSS Selectors Demo</title>
<link rel="stylesheet" href="styles.css" />
</head>
<style>
  /* Element Selector: Styles all instances of the specified elements */
  p {
    color: blue;
    font-size: 18px;
  }

  h2 {
    color: darkred;
    font-size: 24px;
  }

  div {
    background-color: lightgray;
    padding: 10px;
    margin: 10px 0;
  }

  span {
    color: darkblue;
    font-size: 16px;
  }

  h3 {
    color: purple;
    font-size: 20px;
  }

  /* Class Selector: Styles all elements with the specified class */
  .box {
    width: 200px;
    height: 100px;
    background-color: lightgreen;
    text-align: center;
    padding: 20px;
  }

  .text-style {
    font-style: italic;
    color: darkgreen;
  }

  .text-style span {
    color: red;
  }

  /* ID Selector: Styles the element with the specified ID */
  #unique-box {
    width: 250px;
```

```
height: 120px;
background-color: lightcoral;
text-align: center;
padding: 25px;
}

#header-box {
  background-color: lightblue;
}

#main-heading {
  color: navy;
  text-transform: uppercase;
}

#main-paragraph {
  font-size: 16px;
}

#highlighted-span {
  background-color: yellow;
}

/* Descendant Selector: Styles elements inside a parent element */
.container p {
  color: darkgreen;
}

.container span {
  color: orange;
}

.container div {
  background-color: lightblue;
}

.container ul li {
  list-style-type: square;
  color: purple;
}

/* Child Selector: Styles only direct children of the parent */
.parent > p {
  color: green;
}

.parent > span {
  color: brown;
}

.parent > div {
  background-color: lightpink;
}
```

```
.parent > ul > li {
    color: darkred;
}

/* Adjacent Sibling Selector: Styles elements immediately following another element */
h2 + p {
    color: purple;
}

h3 + span {
    background-color: lightyellow;
}

h3 + p {
    color: darkorange;
}

/* General Sibling Selector: Styles all sibling elements following another element */
h2 ~ p {
    color: darkblue;
}

h2 ~ span {
    font-weight: bold;
}

h2 ~ div {
    background-color: lightgreen;
}

/* Group Selector: Applies the same style to multiple elements */
h3,
p,
span,
div {
    border: 1px solid #000;
    padding: 10px;
}

/* Universal Selector: Styles all elements */
* {
    font-family: Arial, sans-serif;
    margin: 5px;
    padding: 5px;
}

/* Attribute Selector: Styles elements based on attributes */
a[target="_blank"] {
    color: red;
    text-decoration: none;
}

a[target="_self"] {
    color: green;
}
```

```
    text-decoration: underline;
}

input[type="text"] {
    background-color: lightyellow;
}

input[type="email"] {
    background-color: lightcyan;
}

img[alt] {
    border: 2px solid black;
}

/* Pseudo-Class Selector: Styles elements based on interaction */
.hover-box:hover {
    background-color: darkblue;
    color: white;
}

a:hover {
    color: darkgreen;
    text-decoration: none;
}

input:focus {
    background-color: lightgray;
}

/* Pseudo-Element Selector: Styles specific parts of elements */
p::first-letter {
    font-size: 200%;
    color: red;
    font-weight: bold;
}

p::first-line {
    font-style: italic;
    color: gray;
}

div::before {
    content: "Start of Div: ";
    color: brown;
    font-weight: bold;
}

div::after {
    content: " - End of Div";
    color: brown;
    font-weight: bold;
}
```

```
span::before {
  content: "Span: ";
  color: darkorange;
}
</style>
<body>
  <h1>CSS Selectors Demo</h1>

  <!-- Element Selector -->
  <p>This is a paragraph styled using an element selector.</p>
  <h2>This is a heading styled using an element selector.</h2>
  <p>Another paragraph styled using the same selector.</p>
  <div>This is a div styled using an element selector.</div>
  <span>This is a span styled using an element selector.</span>

  <!-- Class Selector -->
  <div class="box">This is a box styled using a class selector.</div>
  <div class="box">Another box styled with the same class.</div>
  <p class="text-style">This is a paragraph styled using a class selector.</p>
  <span class="text-style"
    >This is a span styled using a class selector.</span
  >
  <h3 class="text-style">This is a heading styled using a class selector.</h3>

  <!-- ID Selector -->
  <div id="unique-box">This is a unique box styled using an ID selector.</div>
  <div id="header-box">
    This is another box styled with a different ID selector.
  </div>
  <h2 id="main-heading">This is a heading styled using an ID selector.</h2>
  <p id="main-paragraph">This paragraph is styled using an ID selector.</p>
  <span id="highlighted-span">This span is styled using an ID selector.</span>

  <!-- Descendant Selector -->
  <div class="container">
    <p>
      This paragraph is inside a container styled using a descendant selector.
    </p>
    <span
      >This span is inside a container styled using a descendant
      selector.</span
    >
    <div>
      This div is inside a container styled using a descendant selector.
    </div>
    <ul>
      <li>
        This list item is inside a container styled using a descendant
        selector.
      </li>
      <li>Another list item inside a container.</li>
    </ul>
  </div>
```

```

<!-- Child Selector -->
<div class="parent">
  <p>This is a child of the parent div styled using a child selector.</p>
  <span>This is another child of the parent div.</span>
  <div>This div is also a child of the parent div.</div>
  <ul>
    <li>This list item is a child of the ul in the parent div.</li>
  </ul>
</div>

<!-- Adjacent Sibling Selector -->
<h2>Adjacent Sibling Example 1</h2>
<p>This paragraph is adjacent to the heading above it.</p>
<h2>Adjacent Sibling Example 2</h2>
<p>This is another adjacent paragraph.</p>
<h3>Adjacent Sibling Example 3</h3>
<span>This is an adjacent sibling span.</span>
<h3>Adjacent Sibling Example 4</h3>
<p>Another adjacent sibling paragraph.</p>

<!-- General Sibling Selector -->
<h2>General Sibling Example 1</h2>
<p>This paragraph is a general sibling of the heading.</p>
<span>This span is also a general sibling.</span>
<div>This div is another general sibling.</div>
<p>Another paragraph sibling.</p>
<h3>Another heading that has general siblings.</h3>

<!-- Group Selector -->
<h3>Group Selector Example 1</h3>
<p>Paragraph styled with the same group selector.</p>
<span>Span styled with the same group selector.</span>
<div>Div styled with the same group selector.</div>
<p>Another paragraph in the group.</p>
<span>Another span in the group.</span>

<!-- Universal Selector -->
<div>
  <p>This paragraph is styled with the universal selector.</p>
  <span>This span is styled with the universal selector.</span>
  <div>This div is styled with the universal selector.</div>
  <a href="#">This link is styled with the universal selector.</a>
  <ul>
    <li>This list item is styled with the universal selector.</li>
  </ul>
</div>

<!-- Attribute Selector -->
<a href="https://www.example.com" target="_blank">
  >This is a link with target="_blank" styled using an attribute
  selector.</a>
>
<a href="https://www.example2.com" target="_self">
  >This is a link with target="_self" styled using an attribute selector.</a>

```



```
>
<input type="text" placeholder="Text input" />
<input type="email" placeholder="Email input" />


<!-- Pseudo-Class Selector -->
<div class="hover-box">Hover over this box to see the effect.</div>
<div class="hover-box">Hover over another box.</div>
<a href="#">Hover over this link to see the effect.</a>
<input type="text" placeholder="Focus on this input field" />
<input type="text" placeholder="Focus on another input field" />

<!-- Pseudo-Element Selector -->
<p>
  This paragraph demonstrates a pseudo-element. Notice the first letter is
  styled.
</p>
<p>
  Another paragraph where the first line is styled using a pseudo-element.
</p>
<div>This div has a pseudo-element applied to it.</div>
<span>This span has a pseudo-element too.</span>
<h3>This heading demonstrates a pseudo-element.</h3>
</body>
</html>
```